a data science project, particularly the operational ones. Because ML projects are iterative in nature, keeping track of all the variables, configurations, and outcomes may be a challenging task in itself. Effective team cooperation is becoming increasingly important as data science teams get larger and more dispersed.

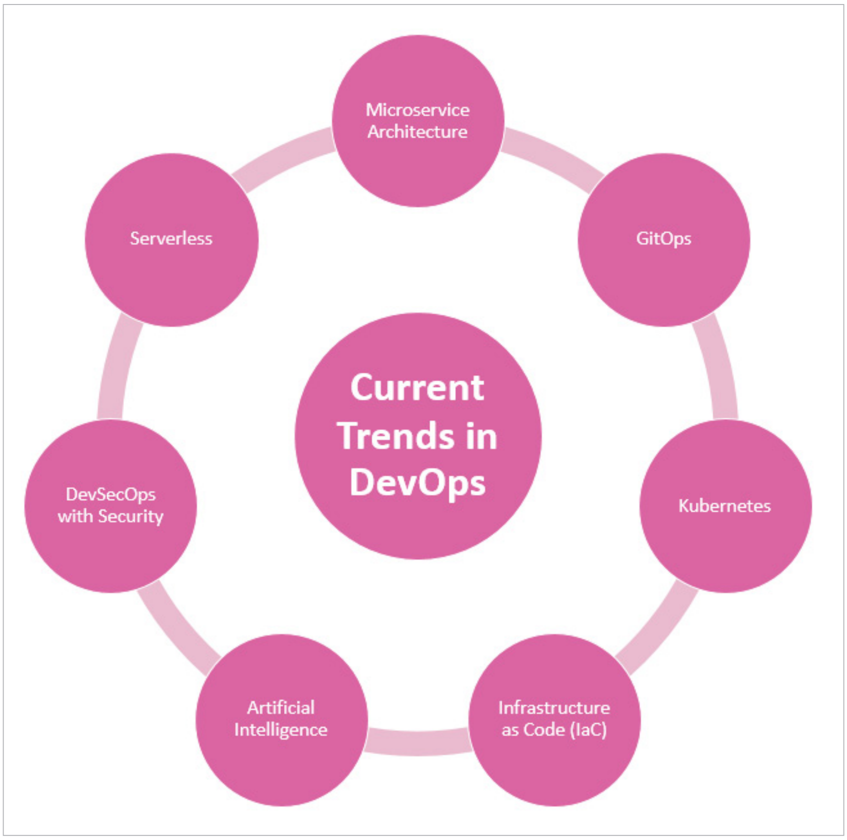

Figure 1: Recent trends in DevOps

Free and open source tools for MLOps

A number of suites and platforms are available for the implementation and deployment of MLOps for different applications. A few key platforms, using which MLOps deployment and overall control on the process becomes easy for project managers and developers, are listed below.

Kubeflow

kubeflow.org

Open source MLOps platform Kubeflow makes orchestrating and deploying machine learning workflows very easy for developers. Machine learning training, pipeline building, and Jupyter notebook management are all supported by Kubeflow's dedicated services and integrations. It's easy to interface with frameworks like Istio, and can even perform TensorFlow training tasks without any issues.

MLflow

mlflow.org

This open source machine learning life cycle management platform provides several components for experiment tracking, project packaging, model deployment, and registry. Machine learning libraries such as TensorFlow and PyTorch may be easily integrated with MLflow to speed up the training, deployment, and maintenance of applications.

Data Version Control (DVC)

dvc.org

An open source tool for data science and machine learning applications, DVC is written in Python programming language. Data sets and machine learning models are managed and versioned in a Git-like fashion. With DVC, machine learning projects may be shared and replicated easily using a command-line interface.

Pachyderm

github.com/pachyderm/pachyderm

Pachyderm is an MLOps data pipeline and versioning leader. You can automate and scale your machine learning life cycle while ensuring reproducibility with its data foundation. It comes in a commercial Pachyderm Enterprise Edition and an open source Pachyderm Community Edition thanks to financing from Benchmark, Microsoft M12, and others. Using Pachyderm, customers can get their ML and AI initiatives off the ground faster, minimise their data processing and storage expenses, and comply with stringent data governance guidelines.

Metaflow

metaflow.org

ML models can be trained, deployed, and managed with ease using Metaflow, a Python based framework that merges machine learning, deep learning, and Big Data. Netflix created Metaflow as an open source MLOps platform that makes it simple to handle large scale enterprise data science initiatives.

Kedro

github.com/quantumblacklabs/kedro

A Python-written open source MLOps framework, Kedro helps data scientists write code that is easy to test, reproduce and maintain. Versioning and modularity are used in machine learning projects as part of the software engineering process. It provides pipeline visualisation, project templates, and the ability to distribute data science projects in a variety of ways.